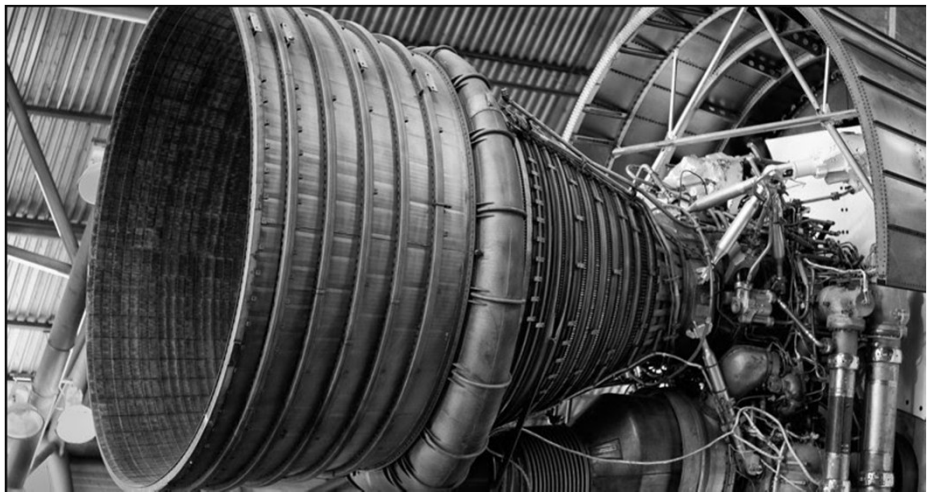
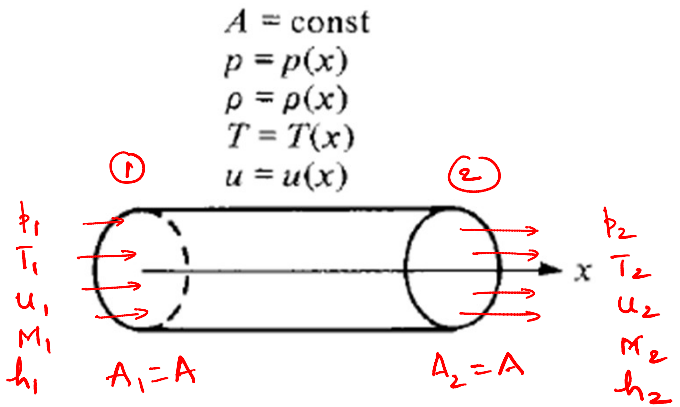
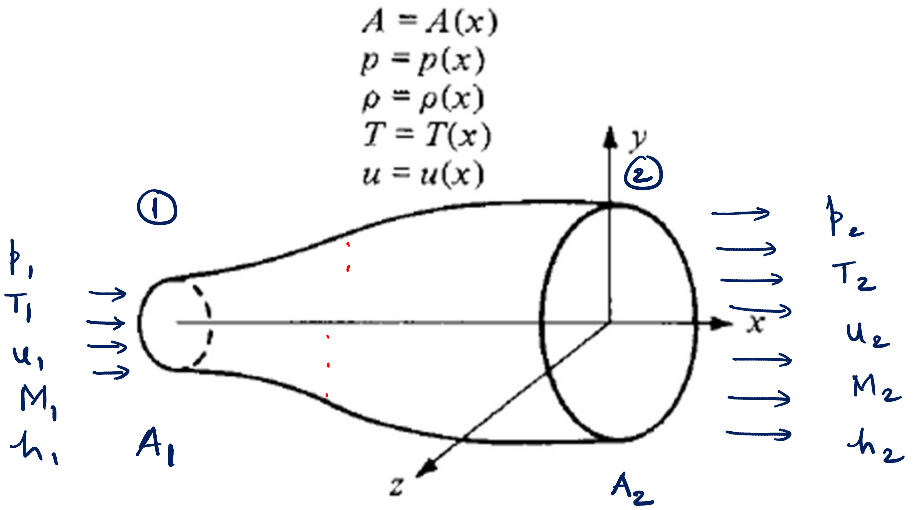
Nozzle flows



One-dimensional flow -> A=constant



Quasi one-dimensional flow -> A=A(x)



Governing equations:

$$\textit{Continuity:} \qquad \rho_1 u_1 = \rho_2 u_2$$

$$\textit{Momentum:} \quad p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$$

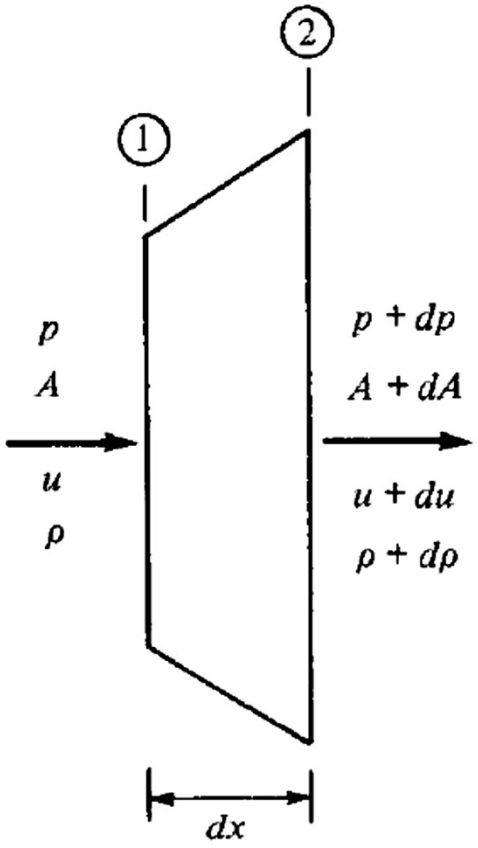
$$\textit{Energy:} \qquad h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

$$\rho_1 u_1 A_1 = \rho_2 u_2 A_2$$

$$p_1 A_1 + \rho_1 u_1^2 A_1 + \int_{A_1}^{A_2} p \, dA = p_2 A_2 + \rho_2 u_2^2 A_2$$

$$h_1 + \frac{u_1^2}{2} = h_2 + \frac{u_2^2}{2}$$

Quasi 1-D Governing equations in differential form



$$Continuity: \quad d(\rho u A) = 0$$

$$Momentum: \quad \frac{dp}{\rho} + u\,du = 0$$ (Compressible Bernoulli eqn)

$$Energy: \quad dh + u\,du = 0$$

Area-Velocity Relation

From continuity equation,

$$d(\rho u A) = 0$$

$$\frac{d\rho}{\rho} + \frac{du}{u} + \frac{dA}{A} = 0 \quad \text{-------- (1)}$$

From momentum equation, $\frac{dp}{\rho} + u\,du = 0$

$$\frac{dp}{d\rho}\frac{d\rho}{\rho} = -u\,du$$

For isentropic flow, $\frac{dp}{d\rho} = a^2$

$$a^2 \frac{d\rho}{\rho} = -u\, du$$

or $$\frac{d\rho}{\rho} = -\frac{u\, du}{a^2} = -\frac{u^2}{a^2}\frac{du}{u}$$

$$\frac{d\rho}{\rho} = -M^2 \frac{du}{u} \quad \text{-------- (2)}$$

Substituting (2) in (1),

$$-M^2 \frac{du}{u} + \frac{du}{u} + \frac{dA}{A} = 0$$

$$\boxed{\frac{dA}{A} = (M^2 - 1)\frac{du}{u}}$$

$$\frac{dA}{A} = (M^2 - 1)\frac{du}{u}$$

$M < 1$

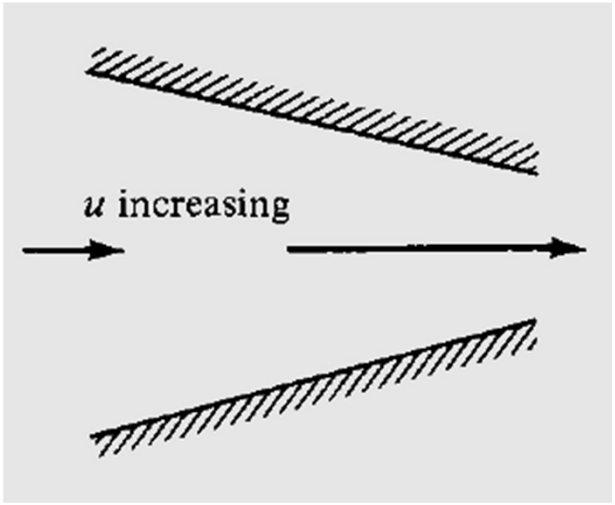
$(M^2 - 1) < 0 \quad (-ve)$

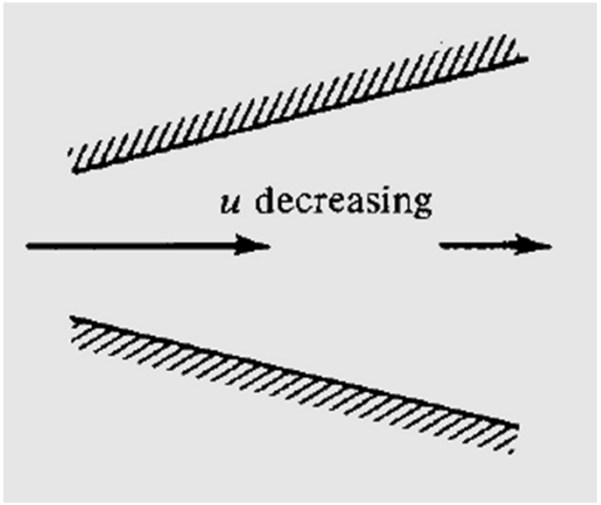
$\frac{dA}{A} = (-ve)\frac{du}{u}$

For $0 \le M < 1$ (subsonic flow), the term $(M^2 - 1) < 0$

- An *increase* in velocity (positive *du*) is associated with a *decrease* in area (negative *dA*).
- Likewise, a decrease in velocity (negative *du*) is associated with an increase in area (positive *dA*)
- For a subsonic compressible flow, to increase the velocity, we must have a convergent duct, and to decrease the velocity, we must have a divergent duct.

$M < 1$





$$\frac{dA}{A} = (M^2 - 1)\frac{du}{u}$$

$M > 1$

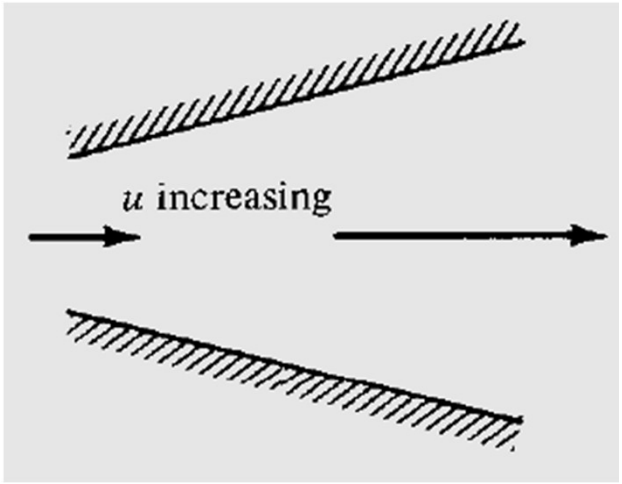
$(M^2 - 1) > 0 \quad (+Ve)$

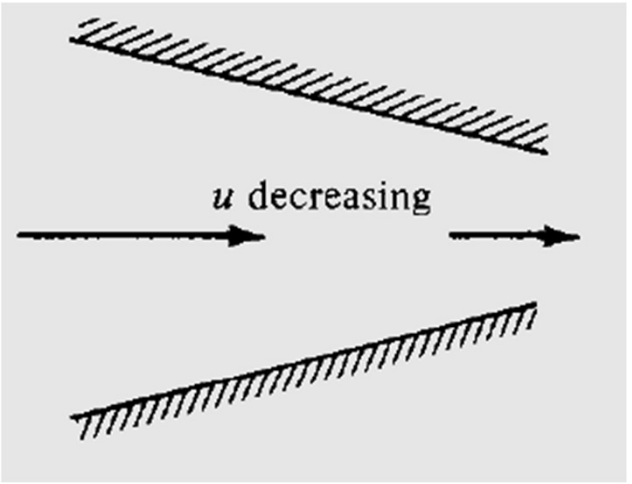
$\frac{dA}{A} = (+Ve)\frac{du}{u}$

For *M* > 1 (supersonic flow), the term $(M^2 - 1) > 0$

- an *increase* in velocity (positive *du*) is associated with an *increase* in area (positive *dA*).
- Likewise, a decrease in velocity (negative *du*) is associated with a decrease in area (negative *dA*).
- For a supersonic flow, to increase the velocity, we must have a divergent duct, and to decrease the velocity, we must have a convergent duct.

$M > 1$





$$\frac{dA}{A} = (M^2 - 1)\frac{du}{u}$$

$M = 1$

$\Rightarrow M^2 - 1 = 0$

$\Rightarrow \frac{dA}{A} = 0 \quad \Rightarrow dA = 0$

For *M* = 1 (sonic flow), the term (M^2 - 1)=0 and therefore d*A*=0

- Mathematically, this corresponds to a local maximum or minimum in the area distribution.
- Physically, it corresponds to a minimum area called **Throat**

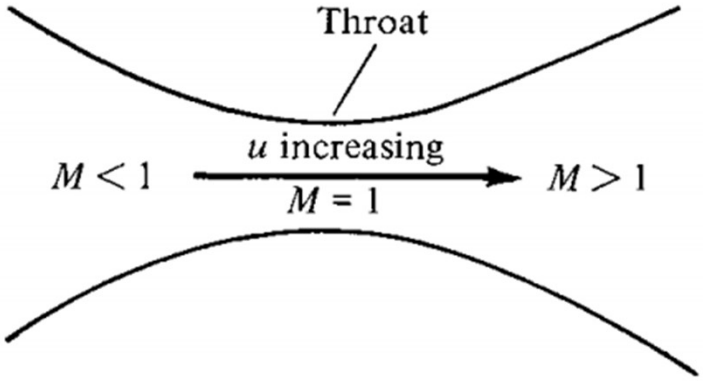
In the limit of M→0 (Incompressible Flow),

$$\frac{dA}{A} = (-1)\frac{du}{u}$$

$$\frac{dA}{A} + \frac{du}{u} = 0 \Rightarrow d(Au) = 0$$

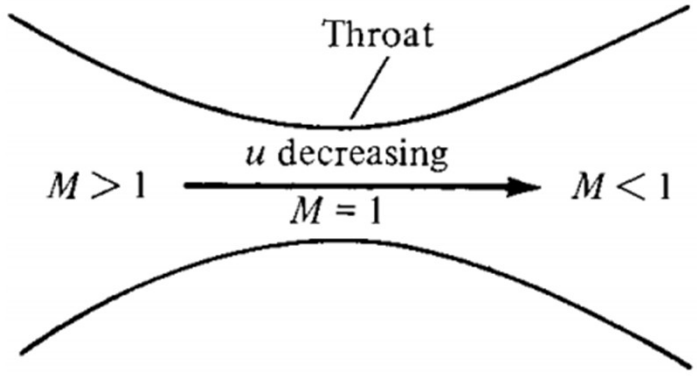
(or) Au = constant

To accelerate a gas at rest to supersonic speeds



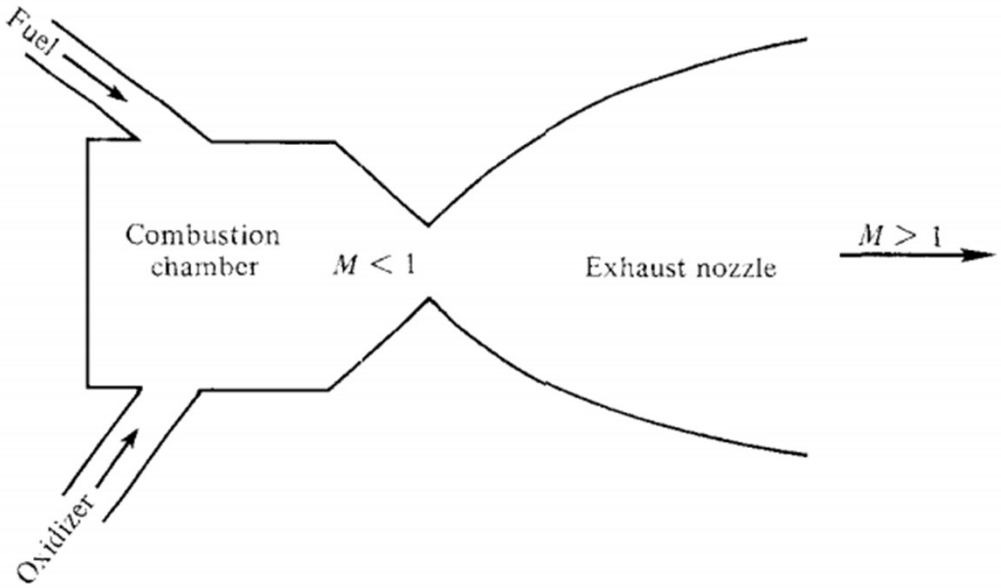
Supersonic nozzle

To slow down a supersonic flow to subsonic

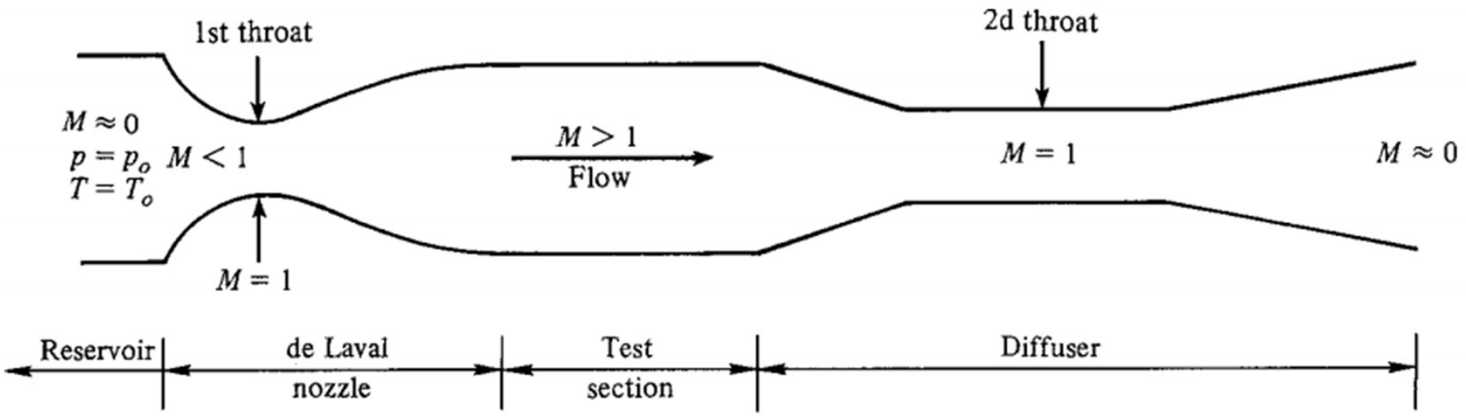


Supersonic diffuser

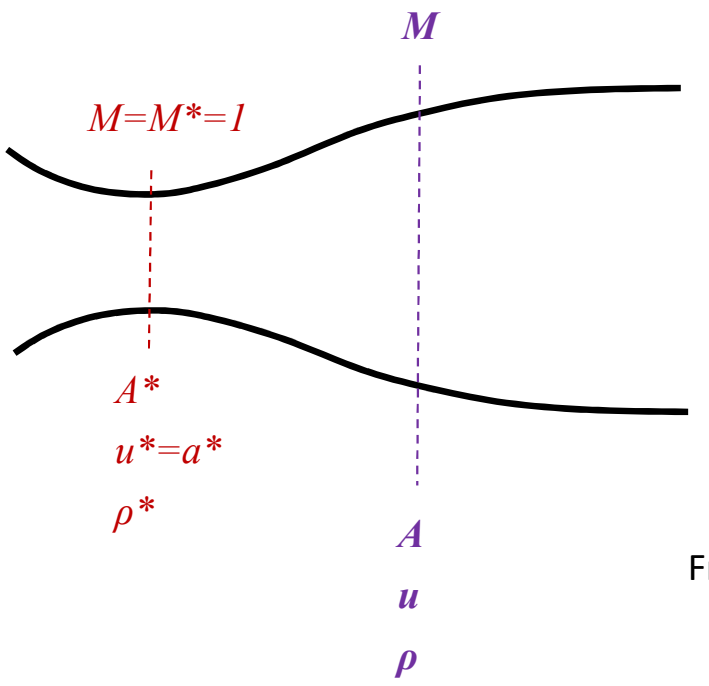
Rocket nozzle



Supersonic wind tunnel



Isentropic flow through a C-D nozzle: Area-Mach number relation



From continuity equation,

$$\rho^* u^* A^* = \rho u A$$

$$\frac{A}{A^*} = \frac{\rho^*}{\rho}\frac{a^*}{u} = \frac{\rho^*}{\rho_0}\frac{\rho_0}{\rho}\frac{a^*}{u}$$

where, ρ_0 is the stagnation density and is constant throughout an isentropic flow.

From isentropic relations,

$$\frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma-1)}$$

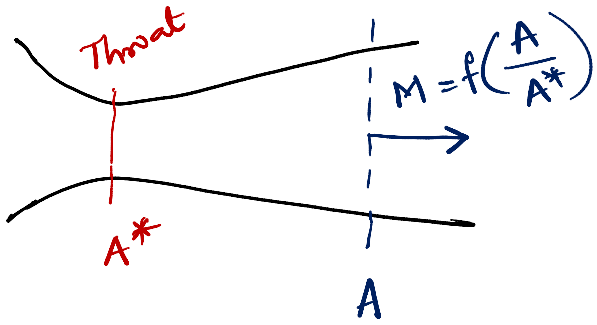
$$\frac{\rho^*}{\rho_0} = \left(\frac{2}{\gamma + 1}\right)^{1/(\gamma-1)}$$

From the definition of M*,

$$\left(\frac{u}{a^*}\right)^2 = M^{*2} = \frac{[(\gamma+1)/2]M^2}{1+[(\gamma-1)/2]M^2}$$

$$\frac{A}{A^*} = \frac{\rho^*}{\rho_0}\frac{\rho_0}{\rho}\frac{a^*}{u} \quad \Longrightarrow \quad \left(\frac{A}{A^*}\right)^2 = \left(\frac{\rho^*}{\rho_o}\right)^2\left(\frac{\rho_o}{\rho}\right)^2\left(\frac{a^*}{u}\right)^2$$

$$\left(\frac{A}{A^*}\right)^2 = \left(\frac{2}{\gamma+1}\right)^{2/(\gamma-1)}\left(1+\frac{\gamma-1}{2}M^2\right)^{2/(\gamma-1)}\left(\frac{1+\frac{\gamma-1}{2}M^2}{\frac{\gamma+1}{2}M^2}\right)$$

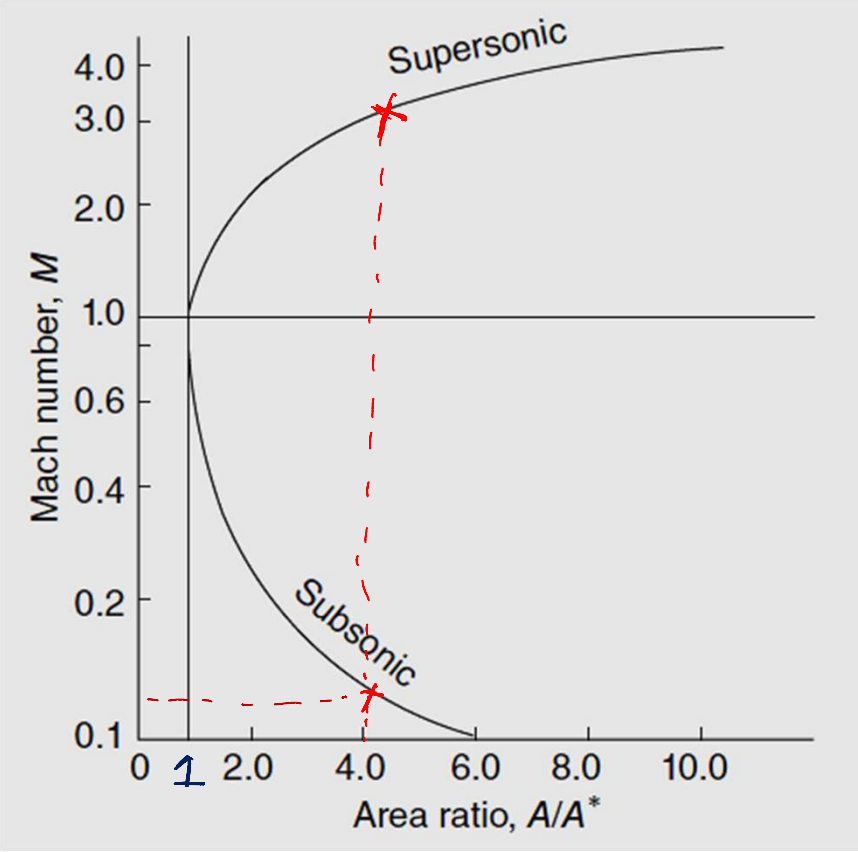


$$\boxed{\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{(\gamma+1)/(\gamma-1)}}$$

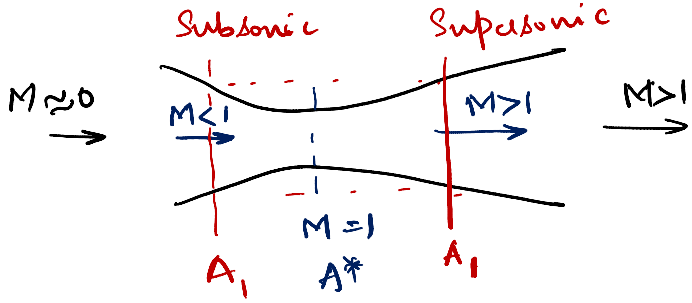
$$M = f(A/A^*)$$

Mach number at any location in the duct is a function of the ratio of the local duct area to the sonic throat area

Area - Mach number relation: $$\left(\frac{A}{A^*}\right)^2 = \frac{1}{M^2}\left[\frac{2}{\gamma+1}\left(1+\frac{\gamma-1}{2}M^2\right)\right]^{(\gamma+1)/(\gamma-1)}$$



- *A* must be greater than or at least equal to *A**; the case where *A* < *A** is physically not possible in an isentropic flow.
- *A/A** ≥ 1 yields *two* solutions for *M* at a given *A/A**—a subsonic value and a supersonic value.
- Which value of *M* that actually holds in a given case depends on the pressures at the inlet and exit of the duct



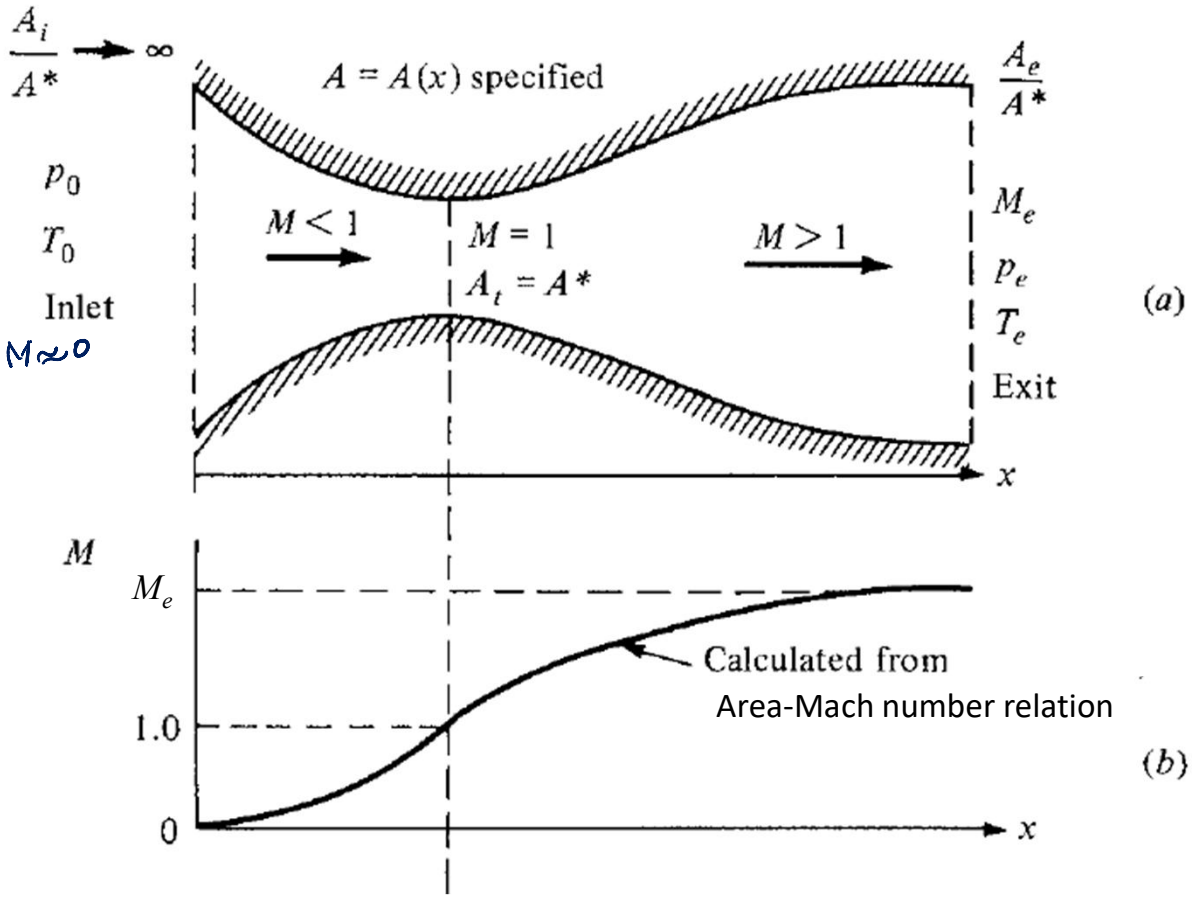
Isentropic flow of perfect gases ($\gamma = 1.4$)

M	M^*	T/T_0	p/p_0	A/A^*	F/F^*	$\frac{A}{A^*}\frac{p}{p_0}$
0.31	0.336	0.981	0.936	1.976	1.655	1.849
0.32	0.347	0.979	0.932	1.922	1.614	1.790
0.33	0.357	0.978	0.927	1.871	1.577	1.735
0.34	0.368	0.977	0.923	1.823	1.542	1.683
0.35	0.378	0.976	0.918	1.778	1.509	1.634

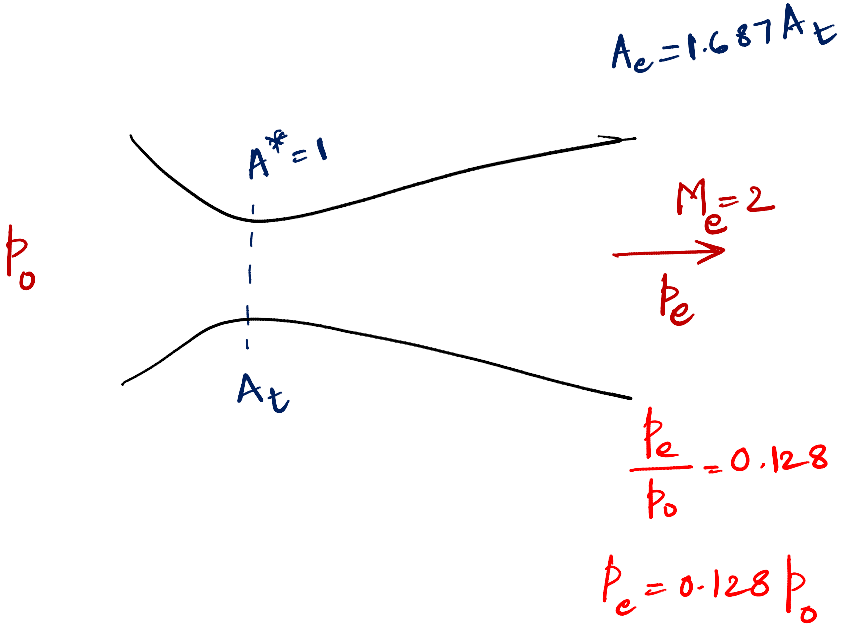
Isentropic flow of perfect gases ($\gamma = 1.4$)

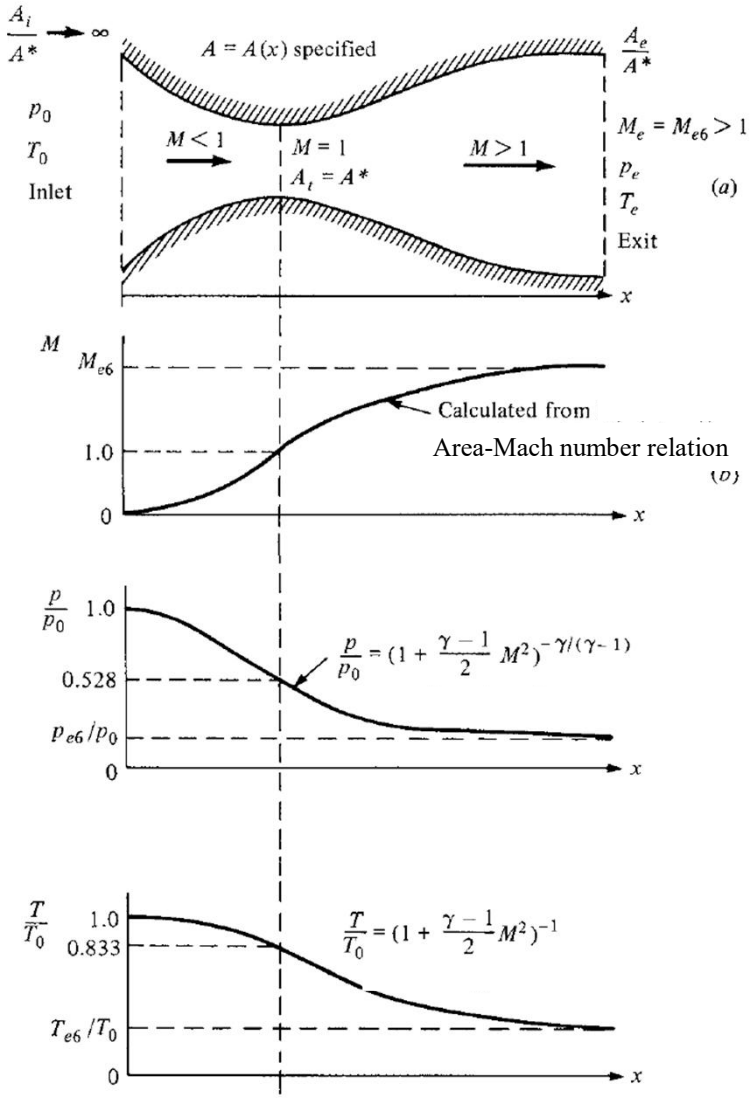
M	M^*	T/T_0	p/p_0	A/A^*	F/F^*	$\frac{A}{A^*}\frac{p}{p_0}$
2.11	1.681	0.529	0.108	1.853	1.138	0.199
2.12	1.685	0.527	0.106	1.869	1.139	0.198
2.13	1.689	0.524	0.1043	1.885	1.141	0.197
2.14	1.694	0.522	0.1027	1.902	1.142	0.195
2.15	1.698	0.519	0.1010	1.918	1.143	0.194

Isentropic supersonic nozzle flow



Isentropic supersonic nozzle flow

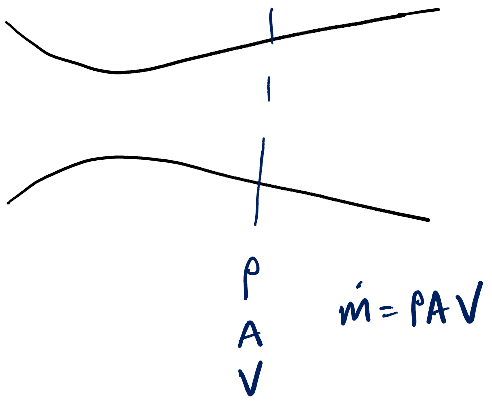




Mass flow rate through a nozzle

Mass flow rate through the nozzle

$\dot{m} = \rho A V$



$$\frac{\dot{m}}{A} = \rho V$$

$$= \frac{\rho}{\rho_0}\rho_0 \frac{V}{a} a$$

$$= \frac{\rho}{\rho_0}\frac{p_0}{RT_0} M\sqrt{\gamma RT}$$

$$= \frac{\rho}{\rho_0}\sqrt{\frac{T}{T_0}}\sqrt{T_0}\frac{p_0 M}{RT_0}\sqrt{\gamma R}$$

Isentropic flow

$$\frac{p_0}{p} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{1}{\gamma-1}}$$

$$\frac{T_0}{T} = \left(1 + \frac{\gamma-1}{2}M^2\right)$$

$$= \frac{p_0 M\sqrt{\gamma/RT_0}}{\left(1+\frac{\gamma-1}{2}M^2\right)^{1/(\gamma-1)}\left(1+\frac{\gamma-1}{2}M^2\right)^{1/2}}$$

$$\frac{\dot{m}}{A} = \boxed{\frac{p_0 M\sqrt{\gamma/RT_0}}{\left(1+\frac{\gamma-1}{2}M^2\right)^{\frac{\gamma+1}{2(\gamma-1)}}}}$$

At what *M* does the mass flow rate becomes maximum ? $\Rightarrow \frac{d\left(\frac{\dot{m}}{A}\right)}{dM} = 0$ Keeping A, P_0, T_0 as constants

- For a specified flow area *A* and stagnation properties T_0 and P_0, the maximum mass flow rate can be determined by differentiating the equation with respect to *M* and setting the result equal to zero. It yields M = 1.
- Since the only location in a nozzle where the Mach number can be unity is the location of minimum flow area (the throat), the mass flow rate through a nozzle is a maximum when M = 1 at the throat.
- This condition (Mach 1 at Throat) is called CHOKED FLOW Condition

Maximum mass flow rate through the nozzle

$$\left(\frac{\dot{m}}{A}\right)_{\max} = \frac{\dot{m}}{A^*} = \boxed{\sqrt{\frac{\gamma}{RT_0}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}\, p_0}$$

$$\dot{m}_{\max} = \boxed{\frac{0.6847}{\sqrt{RT_0}} p_0 A^*}$$

• Choked flow